

1. Administrative & Study Identification

Full Study Title: A Study of the Efficacy and Safety of Lisocabtagene Maraleucel (Liso-cel) as First-Line Therapy in Adults With Transplant-Ineligible Primary Central Nervous System Lymphoma **Short Title/Acronym:** The CAROLYN Trial **Protocol Number:** CA082-1215 **NCT Number:** NCT07015242 **Posting ID:** 122098413411978802 **Sponsor Name:** Bristol Myers Squibb (Juno Therapeutics, Inc., a Bristol-Myers Squibb Company) **Investigational Product:** Lisocabtagene Maraleucel (Liso-cel / Breyanzi) **Phase of Development:** Phase II **Trial Status:** Not Yet Recruiting **Disease Area:** Lymphoma (Primary Central Nervous System Lymphoma) **Medical Monitor:** Bristol-Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the efficacy of lisocabtagene maraleucel (liso-cel) in terms of Progression-free Survival (PFS) as a first-line treatment. **Secondary Objectives:** To evaluate Progression-free Survival (PFS), Modified Progression-free Survival (mPFS), and Complete Response Rate (CRR).

2.2 Background & Rationale To address the high unmet medical need in patients with newly diagnosed Primary Central Nervous System Lymphoma (PCNSL). Up to 30% of patients either do not respond to or experience early relapse with standard front-line chemotherapy. Patients who are ineligible for autologous stem cell transplantation (ASCT) face poor prognoses, making the evaluation of CD19-directed CAR T-cell therapy (liso-cel) as an earlier, first-line option vital to improving clinical outcomes.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Single Group Assignment)

3.2 Planned Enrollment & Duration **Target N\$:** 65 evaluable patients **Number of Sites:** Multicenter (Including regions in the United States and Europe) **Estimated Study Start:** Early 2024 **Estimated Primary Completion:** Not Yet Recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant must be 18 years or older at the time of signing the informed consent form (ICF). 2. Histologically confirmed primary central nervous system (CNS) lymphoma (PCNSL) prior to screening, as assessed by local pathology. 3. Transplant-ineligible based on physician's assessment and meeting at least one of the following criteria: age ≥ 65 years or HCT-CI (Hematopoietic Cell Transplantation-specific Comorbidity Index) score ≥ 3 . 4. Participant must be suitable, per investigator, to receive a high dose methotrexate (HD-MTX) based treatment regimen. 5. Prior to signing ICF, anti-cancer therapy for the treatment of PCNSL must only include standard of care regimens, with or without corticosteroids given for disease-related symptoms. 6. Prior to ICF signature, participant's disease must be sensitive to prior high-dose methotrexate-based regimens, as demonstrated by a complete response (CR, no remaining signs of PCNSL) or a partial response (PR, signs of PCNSL mostly gone) per Investigator's assessment, based on the International Primary CNS Lymphoma Collaborative Group (IPCG) criteria. 7. Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1, or 2. 8. Individuals of childbearing potential (IOCBP) must have a negative highly sensitive pregnancy test within 24 hours prior to the start of study intervention.

4.2 Exclusion Criteria 1. Participant has a diagnosis of secondary CNS lymphoma due to systemic disease. 2. Primary intraocular lymphoma (PIOL) / Primary vitreoretinal lymphoma (PVRL) and isolated cerebrospinal fluid (CSF) disease. 3. Any significant medical condition including the presence of laboratory abnormalities, which places the participant at unacceptable risk if he/she was to participate in the study based on investigator's judgement. 4. History of another primary malignancy that has not been in remission for ≥ 2 years. 5. Prior treatment with CAR T-cell or any other gene therapy product that utilizes human genome-editing technology. 6. History of or active human immunodeficiency virus (HIV). 7. Active hepatitis B or active hepatitis C. 8. Active autoimmune disease requiring immunosuppressive therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: One-time administration of Lisocabtagene Maraleucel (Liso-cel) after required pre-conditioning. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Single infusion with long-term safety and efficacy follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Induction chemotherapy to control disease, followed by Lymphodepleting chemotherapy (Fludarabine and Cyclophosphamide [Cytosan/

Endoxana]). **Prohibited:** Prior CAR T-cell therapy, investigational gene therapies, and chronic immunosuppressive therapies (except as mandated by the study protocol).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Leukapheresis (T-cell collection), Labs, Scans
Pre-conditioning	Day -7 to -2	Lymphodepleting Chemotherapy (Fludarabine + Cyclophosphamide)
Baseline	Day 0	Liso-cel CAR T-cell IV Infusion, Safety Monitoring
Interim	Weeks 4, 8, 12	Safety Labs, Efficacy Assessment (PFS/CRR), Cytokine Release Syndrome (CRS) monitoring
End of Study	Follow-up	Disease progression evaluation, long-term survival follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-free Survival (PFS): Defined as the time from the date of liso-cel infusion to the date of first documented disease relapse or progression as assessed by the investigator, or death from any cause, whichever occurs first. **Measurement Method:** Investigator assessment using International Primary CNS Lymphoma Collaborative Group (IPCG) criteria and regular imaging (MRI).

7.2 Secondary & Exploratory Endpoints 1. Progression-free Survival (PFS) 2. Modified Progression-free Survival (mPFS) 3. Complete Response Rate (CRR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring post-infusion for CAR T-cell specific toxicities, including Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS). **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External, independent data monitoring committee to periodically review safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for patients receiving any fraction of the liso-cel infusion. **Sample Size Justification:** Target sample of N=65 participants powered to detect a statistically significant improvement in PFS over historical standard of care controls.

Handling of Missing Data: Last Observation Carried Forward (LOCF) or censoring at the time of alternative anti-cancer therapy.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-frequency onsite monitoring during the peri-infusion period, transitioning to periodic monitoring during long-term follow-up. **Audit Trail:** Strict documentation for Chain of Identity/Chain of Custody for the patient's cells from leukapheresis through manufacturing to infusion.

11. Study Identifiers & Governance

Primary ID: CA082-1215 **Registry Number:** NCT07015242 **Posting ID:** 122098413411978802 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study of the Efficacy and Safety of Lisocabtagene Maraleucel (Liso-cel) as First-Line Therapy in Adults With Transplant-Ineligible Primary Central Nervous System Lymphoma **Official Regulatory Title:** The CAROLYN Trial: Lisocabtagene Maraleucel as First-Line Therapy for Primary Central Nervous System Lymphoma (PCNSL) in Transplant-Ineligible Patients

12. Clinical Framework (Lymphoma Specific)

Primary Condition: Primary Central Nervous System Lymphoma (PCNSL) **Diagnosis Threshold:** Histologically/cytologically confirmed untreated PCNSL, Transplant-ineligible **Medical Methodology:** Autologous CD19-directed CAR T-cell immunotherapy **Optimization Target:** Achievement of Complete Response (CR) or Partial Response (PR) to bridge to cellular therapy.

13. Study Procedures & Methodology

Management Pathway: Collection of T-cells → Bridging therapy (if needed) → Lymphodepletion → Infusion → Toxicity Management. **Education Protocol (HCP):** Mandated training on the Risk Evaluation and Mitigation Strategy (REMS) program for CRS and neurologic toxicities.

Education Protocol (Patient): Strict guidance on staying near the treatment center for ≥ 4 weeks post-infusion and caregiver support requirements. **Quality Audit Mechanism:** Clinical and manufacturing audits to ensure cell viability and patient safety.

14. Observation & Follow-Up Schedule

Total Duration: Up to 15 years post-infusion (consistent with FDA gene therapy guidelines). **Onsite Visit Cadence:** Intensive monitoring in the first 28 days; followed by Months 3, 6, 9, 12, and then bi-annually. **Remote Monitoring Frequency:** Periodic wellness and survival checks once in the long-term follow-up phase. **Checklist Review Interval:** Regular interval tracking for subsequent anti-lymphoma therapies.

15. Sign-off & Approval

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				